MARK VERSTEGEN, MS, CSCS[1] • SUSAN FALSONE, PT, MS, SCS, ATC, CSCS, COMT[2]
RUSSELL ORR, MS, ATC, CSCS[3] • STEVE SMITH, PT, DPT, SCS, CSCS[4]

Suggestions From the Field for Return to Sports Participation Following Anterior Cruciate Ligament Reconstruction: American Football

Journal of Orthopaedic & Sports Physical Therapy®
Downloaded from www.jospt.org at on October 18, 2016. For personal use only. No other uses without permission.
Copyright © 2012 Journal of Orthopaedic & Sports Physical Therapy®. All rights reserved.

Returning an athlete, specifically an American football player, back to competition after surgery requires a comprehensive approach that, in addition to rehabilitation of the injury, includes nutritional considerations to create an optimal healing environment for the tissue, cardiovascular fitness considerations based on the needs of the athlete, and an awareness of the psychosocial factors that are a part of the injury and recovery process. This comprehensive, integrated approach, requiring multidisciplinary input, bridges the gap between rehabilitation and performance. There is no one person who can do everything for the athlete as he progresses from the operating room to return to sport performance on the field. There are many healthcare professionals involved in the process, some playing larger roles at certain times than others. In the athlete-centered model, everyone is involved in the process, working together to return the athlete to play.

SYNOPSIS: Returning an American football player to sport after an anterior cruciate ligament reconstruction is challenging on several fronts. First, there are approximately 15 different positions a football player could play, depending on how specifically you define the positions on the field. Each of these positions has specific demands for optimal size, strength, power, body composition, cardiovascular fitness, and movement. Understanding all of these factors is paramount to returning a football player not only to his sport but also to his specific position. Second, the chaotic, contact-rich nature of the sport requires that heavy demands be placed on the lower extremities to attenuate external contact forces from other players and from ground reaction forces associated with accelerating, decelerating, quick stops and starts, and changing direction. Finally, return to a competitive level of performance is further influenced by playing surface, shoe selection, the equipment the player wears, and various potential psychosocial factors. It is the responsibility of the clinician to provide a progressive and systematic rehabilitation program by first introducing preprogrammed movements that, once mastered, are progressed with elements of more reactive and random movement patterns. The precise nature of this systematic progression of advanced rehabilitation is key in developing return-to-play criteria and, ultimately, in readying an American football player for eventual return to sport and a particular position.

LEVEL OF EVIDENCE: Therapy, level 5.
J Orthop Sports Phys Ther 2012;42(4):337-344.
doi:10.2519/jospt.2012.4031

KEY WORDS: *ACL, criteria, rehabilitation, surgery*

Because the focus of this manuscript is on late-stage functional rehabilitation after anterior cruciate ligament (ACL) reconstruction in an American football player, we will not cover the early stages of rehabilitation. The criteria for entering the functional rehabilitation phase can include but are not limited to the following: full pain-free knee range of motion, a minimum level of quadriceps strength,[12,17] no episodes of giving way and a negative pivot shift test upon examination, no significant increases in edema seen with standard rehabilitation, a minimum knee function questionnaire score,[17] and a Functional Movement Screen (FMS) test score of 14 or higher, without asymmetries or pain. We are strongly in favor of using the FMS to determine an athlete's readiness to begin the functional stage of

[1]Founder and President, Athletes' Performance and Core Performance, Phoenix, AZ. [2]Vice President, Performance Physical Therapy and Team Sports, Athletes' Performance and Core Performance, Phoenix, AZ; Head Athletic Trainer/Physical Therapist, Los Angeles Dodgers, Los Angeles, CA. [3]Director of Tactical Performance, Athletes' Performance Florida, The Andrews Institute, Gulf Breeze, FL. [4]Rehabilitation Coordinator, Los Angeles Dodgers, Phoenix, AZ; Performance Physical Therapist, Athletes' Performance, Phoenix, AZ. Athletes' Performance uses the Functional Movement Screen in its methodology. Dr Smith is also an instructor for Functional Movement Systems. The Functional Movement Screen and the Y-Balance Test are components of the system. Address correspondence to Dr Steve Smith, Rehabilitation Coordinator, Los Angeles Dodgers, Camelback Ranch, Phoenix, AZ 85037. E-mail: ssmith@athletesperformance.com or ssmith@ladodgers.com

rehabilitation, as evidence suggests that it is a useful tool to predict injury risk in multiple populations[5,13,29] and it allows one to see if satisfactory results on measures of impairment at a specific joint correlate with multisegmental movement competency and adequate motor control before the body is under higher stress in an advanced stage of rehabilitation. For similar reasons, a tool like the Y Balance Test could be used to determine if asymmetry exists in the weight-bearing function of the lower extremities.[26] The approximate amount of time after an ACL reconstruction that someone would enter into the functional rehabilitation stage varies greatly and is dependent on age of the patient, graft choice, concomitant surgical procedures, other coexisting injuries, and preference for use of a criteria- or time-based postoperative progression. In general, we feel that this phase should start somewhere around 4 months postsurgery.

SPORT-SPECIFIC REHABILITATION PHASES

There may be several potential phases in a sport-specific functional rehabilitation program that are neither finite nor definitive. Functional rehabilitation is not a true linear progression, as a rehabilitating player may often work on acquiring several movement competencies and multiple goals at the same time. The functional rehabilitation program emphasizes the achievement of a high level of movement quality with competent skill performance before progressing to the next, more complex phase.

We recommend that phase creation be based on concepts of motor learning, with attention to the necessary movement skills needed for the position of the player. Phase progression should move from simple to complex. In each phase, movements and motor qualities that will ultimately be used in competition are the focus. Parameters, instruction, and exercise/drill sets are dictated by the athlete's level of motor learning, movement quality, proprioceptive capabilities, and tissue tolerance. We have 3 phases for our functional rehabilitation program.

TABLE 1 Initial Phase of Functional Rehabilitation (online videos 1 and 2)

Movement Term	Definition	Example
Discrete skill	Skill or task organized in a way that the action is usually brief and has a well-defined beginning and end	10 yd (10 m) of linear acceleration
Closed skill	Skill performed in an environment that is predictable or stationary and that allows performers to plan their movements in advance	Performing 10 yd (10 m) of linear acceleration with the athlete's self-determined start or in response to a predetermined visual or verbal cue from a coach or rehabilitation professional
Blocked practice	Practice sequence in which individuals rehearse the same skill repeatedly	Repeating 10 yd (10 m) of linear acceleration for 10 consecutive repetitions before moving on to the next drill/skill

TABLE 2 Final Phase of Functional Rehabilitation

Movement Term	Definition	Example
Serial task	Several discrete actions connected together in sequence, often with the order of the actions being crucial to performance success	Linking of linear acceleration, linear deceleration, and an outside leg cut to the right or left
Open skill	Skill performed in an environment that is unpredictable or changing around the performer and that requires the performer to adapt their movements in response to dynamic properties of the environment	Performing/linking several discrete actions in response to another player's movements or in response to a previously undetermined cue given by a coach or rehabilitation professional
Random practice	Practice sequence in which individuals perform a number of skills in a (quasi) random order, thus avoiding or minimizing consecutive repetitions of any single skill	Randomly changing the other player's movements or cues given so that the athlete has to vary their movement response accordingly without repeating the same skill

Initial Phase: Discrete, Closed Movements in a Blocked-Practice Format

Discrete movements are defined as movements that have a recognizable beginning and end. Closed movements are predictable (**TABLE 1**, **ONLINE VIDEOS 1** and **2**). The athlete has a fixed stimulus or set of instructions, as directed by the healthcare professional or the athlete's own pace. Blocked practice, although not proven to improve skill performance or transference in a game situation, is the performance of 1 particular skill for a certain number of repetitions before moving on to the next. The blocked-practice format is an appropriate place to start. Examples of exercises/drills in this phase would be 3 sets of 5 countermovements, single-leg hurdle hops, 30-second linear jogging intervals on a treadmill at a specific speed and grade, or 5 repetitions of resisted acceleration from a 2-point start.

Intermediate Phase: Combination Phase

In this phase, combinations of the movement and motor-learning components of the initial and final phases of the program are used (**ONLINE VIDEOS 1** and **2**). The subphases and drills performed in this phase, like those of the initial and

Journal of Orthopaedic & Sports Physical Therapy®
Downloaded from www.jospt.org at on October 18, 2016. For personal use only. No other uses without permission.
Copyright © 2012 Journal of Orthopaedic & Sports Physical Therapy®. All rights reserved.

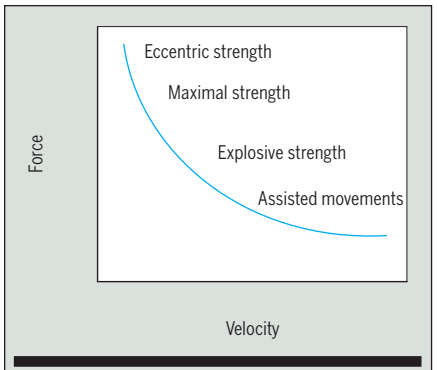


FIGURE 1. Force-velocity curve (strength types plotted on curve). Adapted from Siff and Verkhoshansky[33] and Poliquin and Patterson.[27]

TABLE 3 PLYOMETRIC SESSION DESIGN

Variable	Recommendation
Training frequency/duration of session	2 to 4 times per wk, 10 to 15 min
Movements per session	2 to 3
Sets/repetitions per session	4 to 6 sets, 5 to 8 repetitions
Rest between sets	30 to 90 s
Daily/weekly amount	25 to 50 foot contacts per session, <120 foot contacts per wk

final phases, are at the discretion of the healthcare professional. Examples of exercises/drills in this phase would be a timed T-test, L-drill, 5-10-5/pro agility shuttle, in which several movement skills are linked together but in a predictable order. In these examples, the drill is serial in nature, but the environment is closed and the practice format could be either blocked or random.

Final Phase: Serial, Open Movements in a Random-Practice Format

Serial movements are defined as those that are made up of a series of individual movements linked together (**TABLE 2**). Open movements occur in an environment that is constantly changing and therefore require constant adjustment by the athlete. Random practice is when the athlete practices a number of different tasks in no particular order. Examples of exercises/drills in this phase would be when the player, potentially in a position-specific manner, reacts to another player's movements or the verbal and/or visual cues from a coach or rehabilitation professional. The individual movements are linked together in a changing environment and can be randomly practiced.

Typical and effective rehabilitation focuses on regaining neuromuscular control, reinforcing a movement pattern, and building strength in an atrophied muscle group or limb. It is very typical to see a set/repetition scheme that generally uses 3 sets of 10 repetitions. The rehabilitation specialist must understand, however, that the athlete does not function in a world of 3 sets of 10. Athletes function in a world of varying intensities, varying speeds, and unpredictability. They often must overcome not only the force they generate but also the forces others place on them to the limit of their strength, speed, and range of motion. Movement skills, like rehabilitation phases, must prepare athletes not only to work in their own preplanned motor program but to respond to stimuli that they would not encounter under normal, nonsport playing circumstances. Therefore, sports rehabilitation in this advanced stage must provide varying speeds, intensities, volume, and external forces to prepare the athlete to return to the playing field. A simple force-velocity curve (**FIGURE 1**) is useful when developing exercises/drills that provide these various stimuli. An understanding of the different types of strength, the capabilities of the neuromuscular system, and how relevant certain types of strength are to a particular position is also essential.[27] For example, offensive and defensive linemen may need more exercises/drills that elicit more force production and relatively less velocity of movement, while skill-position players may need more exercises/drills that elicit more velocity of movement and relatively less force production.

Considering the multitude of position-specific movements a football player has to perform, exercise and drill selection, like rehabilitation phases, must progress from simple to complex. Each position-specific movement can be broken down into its component parts, and the components can be taught individually before being combined. These complex movements, when broken into their component parts, can be classified into 3 groups of movement possibilities: linear movement, multidirectional movement, and jumping/landing. Once these movements are mastered individually, they can be put together to create more complex movement patterns.

Linear Movement Linear movement consists of moving in 1 direction (**ONLINE VIDEO 1**). An athlete accelerates, runs at top speed, or decelerates. With the exception of track runners and marathoners, the majority of athletes do not solely function in a linear movement pattern. Therefore, deceleration is a transitional movement, as the way an athlete decelerates will often depend on the next movement to be performed. Eccentric strength in both double- and single-leg support is vital when working generally on deceleration. An athlete must master these movement patterns individually prior to putting them together into more complex movement patterns.

Multidirectional Movement Multidirectional skills traditionally consist of lateral movements, such as shuffle patterns, that combine change of direction through transitional movement skills such as crossover and drop-step techniques. Generally, multidirectional movements are used as transitional movements from one pattern to another and are preceded by plant, cut, pivot, or other rotational movements (**ONLINE VIDEO 2**). For example, a defensive back covering a wide receiver in American football may backpedal

Journal of Orthopaedic & Sports Physical Therapy®
Downloaded from www.jospt.org at on October 18, 2016. For personal use only. No other uses without permission.
Copyright © 2012 Journal of Orthopaedic & Sports Physical Therapy®. All rights reserved.

TABLE 4 Drill Progression Through the Functional Rehabilitation Phases: Offensive Lineman*

Phase	Setup and Cues
1. Initial phase (**ONLINE VIDEO 1**)	Perform certain number of repetitions of same skill consecutively at predetermined distance with a clear beginning and end
	Fixed loading style: free, resisted, or assisted
	Perform each repetition with player fully rested
	Perform individual movement skills in isolation
2. Intermediate phase (**ONLINE VIDEO 1**)	Perform certain number of repetitions of similar movement skills
	Create drills linking 2 or more movement skills together
	Begin varying environment and/or athlete movement demands (ie, visual, auditory, reactive, tactile, outcome)
	Examples: change/vary distances of individual movement skills, change work-rest ratios, use a varied loading style with emphasis placed on resisted loading to increase force production, introduce preprogrammed contact
3. Final phase	Perform terminal movement skill repetitions in a randomized order
	Vary environment and/or athlete movement demand (ie, visual, auditory, reactive, tactile, outcome)
	Examples (in addition to those noted above): random contact, reaction to an offensive player's movements, with equipment

**Terminal position-specific movement desired: run block, pull right or left. Component movement skills: drop-step, acceleration, outside leg cut, acceleration downfield.*

TABLE 5 Drill Progression Through the Functional Rehabilitation Phases: Defensive Back*

Phase	Setup and Cues
1. Initial phase (**ONLINE VIDEO 2**)	Perform certain number of repetitions of same skill consecutively at predetermined distance with a clear beginning and end
	Fixed loading style: free, resisted, or assisted
	Perform each repetition with player fully rested
	Perform individual movement skills in isolation
2. Intermediate phase (**ONLINE VIDEO 2**)	Perform certain number of repetitions of similar movement skills
	Create drills linking 2 or more movement skills together
	Begin varying environment and/or athlete movement demands (ie, visual, auditory, reactive, tactile, outcome)
	Examples: change/vary distances of individual movement skills, change work-rest ratios, use a varied loading style with emphasis placed on assisted loading to increase velocity of movement, introduce preprogrammed contact
3. Final phase	Perform terminal movement skill repetitions in a randomized order
	Vary environment and/or athlete movement demand (ie, visual, auditory, reactive, tactile, outcome)
	Examples (in addition to those noted above): random contact, reaction to a defensive player's movements, with equipment, reaction to a thrown ball

**Terminal position-specific movement desired: pass coverage, man to man. Component movement skills: backpedal, 45° drop-step, linear acceleration.*

and incorporate a pivot into a rotational drop-step technique to transition into a linear-acceleration pattern. The drop-step technique is used as a transitional movement to change from a backward direction to a forward direction. An athlete must master many movement patterns and transitions individually prior to integrating them into more complex movement patterns.

Jumping and Landing An athlete must be able to do both of these movements. The intent of plyometric training is to increase the athlete's ability to handle rapid-stretch loads within the musculotendinous unit during a rapid change of direction, mainly during rotational and deceleratory tasks. Plyometrics are executed with double-leg support (jump), single-leg support (hop), or alternating-leg support (bound). These movements can be done in place or in various vectors or planes (linear/sagittal, lateral-medial/frontal, rotational/transverse), with emphasis on vertical or horizontal displacement. Special emphasis is placed on limb alignment and force attenuation during the landing phase. A progressive program is developed by manipulating amplitude, frequency, contact volume, contact linking, and exposure to the stretch-shortening cycle. **TABLE 3** shows an example of a plyometric session design.

Specific/Unique Drills in a Given Phase

TABLES 4 and **5** show 2 examples of how an exercise/drill, knowing the terminal skill goal, can be manipulated and progressed through all the phases of the functional rehabilitation program for a lineman and a skill-position player, respectively. Drill selection and progression for each position are based on all the preceding background information provided.

TABLE 6 shows an example of how different types of strength qualities, the force-velocity curve, the movement type, and program phase interact to create the drills that are prescribed to the recovering athlete at a specific point in the functional rehabilitation process.

Journal of Orthopaedic & Sports Physical Therapy®
Downloaded from www.jospt.org at on October 18, 2016. For personal use only. No other uses without permission.
Copyright © 2012 Journal of Orthopaedic & Sports Physical Therapy®. All rights reserved.

RETURN-TO-FOOTBALL CRITERIA

In the current literature, there is a great amount of variability in the criteria selected by sports medicine professionals to ascertain whether an athlete is ready to return to the team.[15] Common criteria include subjective, clinical, and functional components. The absence of pain, the absence of swelling before, during, and after activity, and self-reports of no episodes of giving way are critical criteria to be met prior to returning to American football. In addition, favorable scores on self-reported measures of knee function and disability are used.[31,37] Likewise, clinical components have included measures of anterior laxity using arthrometry devices,[22,32,35] the absence of a pivot shift upon clinical examination, and a minimum level of proficiency when strength of quadriceps and hamstrings is tested and bilateral comparisons are made.[17,37] Various types of functional tests, including hop and agility tests, have also been recommended[8,17,18,21,34] as a way to assess dynamic knee stability. There is conflicting evidence on how well subjective or clinical assessment outcomes correlate to functional tests.[1,16,20,25,30,32,35,37] Consequently, conducting a battery of tests is generally recommended.[6,36]

The design of any return-to-play test or battery of tests should allow the clinician to answer the following question: "Have I minimized all of the potential risk factors for reinjury when my athlete returns to performing his football skill?" It is our responsibility as rehabilitation specialists to identify and minimize all the risk factors that we can and to ensure adequate tissue tolerance. We must frame the return-to-play question in this context because we are able to affect certain variables that have been shown in the literature to be risk factors for injury. Prospective studies have shown that a history of previous injury continues overwhelmingly to lead all other risk factors for potential future injury.[7,23,24] Another risk factor for injury is asymmetry.[14,18,19,26]

TABLE 6 Example of an Initial-Phase Exercise/Drill Prescription

Strength/Movement	Goal	Linear Acceleration (ONLINE VIDEO 1)	Multidirectional: Change of Direction (ONLINE VIDEO 2)
Eccentric strength	Maximize relative leg strength in all major muscle groups and planes of motion eccentrically for deceleration	Resisted walking lunge	Bungee-assisted lateral cut
Maximum strength	Maximize relative leg strength in all major muscle groups and planes of motion	Rearfoot-elevated split squat	Resisted lateral load and lift wall drill
Starting strength	To overcome inertia in a controlled manner	Box blast: single	Crossover with bullet belt
Explosive strength	To produce a certain amount of work in a defined period of time. To progress, decrease time while doing same amount of work	Hang clean	Drop-step to acceleration
Resisted strength	To control body against an outside force. To be able to stabilize against a force being applied to the body or limb	Resisted sled acceleration runs/starts	Crossover march with sled resistance
Reactive movements	To control body while reacting to someone else's movement, verbal command, or visual command	Get-up start (verbal)	Mirror drill (visual)
Free movements	To have the ability to execute a preprogrammed motor plan	Falling start	6-yd (6-m) hitch right or left
Assisted movements	To control the body movements while an outside source pushes the body beyond normal speed or movement patterns	Bungee assisted (5 yd [5 m])	Bungee-assisted change of direction in athletic base (3 yd [3 m])

Asymmetries that need to be considered include, among others, mobility, stability, recruitment patterns, strength, and movement patterns. Altered motor control[38] and body mass index are also potential risk factors. However, it is our belief that in this population a multisite body composition measurement may be a better marker than body mass index, because the former is an actual site-specific estimate of lean mass and fat mass, while the latter is only a correlation between height and weight.

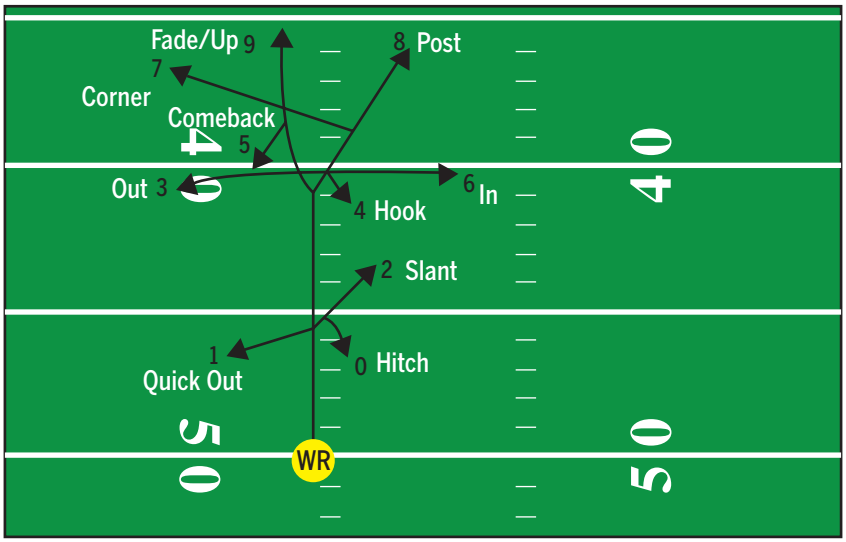
Outcomes or Functional Tests to Clear Return to Play

A study of elite-level running backs and wide receivers with ACL reconstructions showed that these athletes generally returned to their first game at 9 to 12 months following surgery.[3] For the purpose of this paper, "return to football" will be defined as the point at which an athlete is ready to return to individual and team practice. Based on clinical experience, we feel that this milestone should be met 1 to 2 months prior to the athlete's return to a competitive game. This means that our rehabilitation efforts need to focus on returning players to practice and not just up to the point of being able to performance train.

To determine if an athlete is ready to return to any movement-based sport, es-

Journal of Orthopaedic & Sports Physical Therapy®
Downloaded from www.jospt.org at on October 18, 2016. For personal use only. No other uses without permission.
Copyright © 2012 Journal of Orthopaedic & Sports Physical Therapy®. All rights reserved.

pecially one that is chaotic and involves violent collisions, the criteria or tests that are used should replicate the conditions of the sport and position as closely as possible. The criteria used to assess readiness should include position-specific, random movements that allow for bilateral comparison. Movement-based testing should be done with the athlete in a stressed or fatigued state. Because fatigue adversely affects motor control,[11] it is important to see these potential problems in a controlled environment before placing the athlete back into such conditions under more random circumstances. Giving special consideration to fatigue is supported by the data from Bradley and colleagues,[2] who reported that the majority of ACL injuries in the National Football League (NFL) that happened during games occurred later in the game. The work-rest ratios used during testing should also be similar to those occurring during American football. Work done by Iosia and Bishop,[10] as well as Hitchcock and colleagues,[9] provides insight into the work-rest ratios and exercise intensity that should be used to properly fatigue a football player in a sport-specific manner immediately before testing. Work-rest intervals for the fatigue protocol should be set at approximately 5 to 10 seconds of work to 40 seconds of rest. Also, during the work intervals, the intensity of the activity should be high, due to the fact that football players rely primarily on an anaerobic source of energy for adenosine triphosphate (ATP) resynthesis. We recommend that maximum heart rate, recovery heart rate, number of repetitions, and wattage (given access to adequate equipment) are recorded. Once the athlete is fatigued, a recovery period is given, and then the randomly ordered, position-specific movements are initiated. Form breakdown and/or a drop in speed/power are monitored as they were during the fatigue protocol, and recovery time is given as needed. The randomly ordered, position-specific movements are continued until all are completed. An active cool-down period is given following completion. A detailed example of the testing protocol for a wide receiver, including the fatigue protocol, can be found in **FIGURE 2**.



Position-Specific Movements: Wide Receiver

0. 6-y hitch, right and left
1. 6-y out, right and left
2. 3-step slant, right and left
3. 10- to 12-y speed out, right and left
4. 10- to 14-y hook, right and left
5. 18-y comeback, right and left
6. 14- to 16-y square in, right and left
7. 7-step post-corner, right and left
8. 7-step post, right and left
9. 20- to 25-y fade, right and left

Example of a Return-to-Play Testing Protocol: Wide Receiver

1. 5-minute dynamic warm-up
2. 5- to 10-second high-intensity work intervals with 40-second rest (use a piece of equipment incorporating the entire body), until speed or power drops and/or form breaks down
3. Recovery period where heart rate gets to 65% to 70% of maximum or for 2 to 3 minutes
4. Initiate random, position-specific movements (bilaterally compare each movement in a random order), giving 40-second rest between position-specific patterns, until speed or power drops off and/or form breaks down
5. Repeat step 3 as needed
6. Repeat step 4 until all patterns are completed
7. 5-minute active cool-down

FIGURE 2. Example of a return-to-play testing protocol for a wide receiver. One yd equals 0.91 m. Route distances are not to scale on the figure. Abbreviations: L, left; R, right.

The position-specific movements are assessed for a comparison of the time it takes to complete the bilateral pattern and a comparison of the control and ease exhibited during transitional movements. We want to observe and note any movements that may cause an ACL tear or that may excessively stress the healing graft. Specifically, we are looking for the absence of excessive sagittal plane shear of the tibiofemoral joint (seen with increased tibial angle upon weight acceptance) and the absence of tibiofemoral rotation stress to the tibiofemoral joint (seen with femoral medial rotation/collapse or a pronatory foot collapse with tibial rotation).[28] We also recommend a comparison of the first step, stride

Journal of Orthopaedic & Sports Physical Therapy®
Downloaded from www.jospt.org at on October 18, 2016. For personal use only. No other uses without permission.
Copyright © 2012 Journal of Orthopaedic & Sports Physical Therapy®. All rights reserved.

length, and stride frequency bilaterally. Subjective reports of pain or posttesting swelling also need to be taken into consideration.

In addition, to more accurately judge the quality of the athlete's motor control and quantity of the movement skill displayed for asymmetry, we recommend using some type of system to capture kinematic as well as kinetic data.[4] If these systems are not available, a skilled eye that has experience observing various movement patterns will suffice. Eventually, having movement-based qualitative standards for performance and skill-level tasks would also be helpful, just as the FMS is currently used to screen and rank dysfunctional and/or painful fundamental movement. Last, we also see the potential benefit of adding reactions to a thrown football and/or performing the tests in full pads, shells, and/or cleats. These could be used in more advanced testing and could potentially show asymmetries and altered motor control because of the increased difficulty and energy costs associated with the additions.

Prevention of Injury After ACL Reconstruction

Because a history of previous injury is the biggest risk factor for future injury, American football players who have undergone ACL reconstruction are more likely to be injured in the future. To lessen the risk, a high level of movement competency must be maintained through all stages of rehabilitation. Therefore, the athlete must maintain a stable and mobile platform from which to move (a high level of "pillar strength"), so that excessive external forces inherent to the game do not unexpectedly create macrotrauma or even repetitive chronic microtrauma. One area that can be easily influenced is the athlete's warm-up for both training and competition. We recommend that the warm-up be dynamic, weight bearing, proprioceptively rich, activate key stabilizers, and actively elongate/stretch the myofascial system.

CONCLUSION

The task of returning any athlete to sport is both rewarding and challenging, and returning an American football player to sport is no different. It is best to treat this process as a continuum of care from surgery to return to play. The functional or late-stage portion of the rehabilitation process is not a simple, linear progression but rather a well-thought-out mix of art and science. Any healthcare professional involved in this process should always have a working knowledge of the position-specific movement patterns and metabolic characteristics of the athlete's sport. ◉

REFERENCES

1. Bach BR, Jr., Jones GT, Hager CA, Sweet FA, Luergans S. Arthrometric results of arthroscopically assisted anterior cruciate ligament reconstruction using autograft patellar tendon substitution. *Am J Sports Med*. 1995;23:179-185.
2. Bradley JP, Klimkiewicz JJ, Rytel MJ, Powell JW. Anterior cruciate ligament injuries in the National Football League: epidemiology and current treatment trends among team physicians. *Arthroscopy*. 2002;18:502-509. http://dx.doi.org/10.1053/jars.2002.30649
3. Carey JL, Huffman GR, Parekh SG, Sennett BJ. Outcomes of anterior cruciate ligament injuries to running backs and wide receivers in the National Football League. *Am J Sports Med*. 2006;34:1911-1917. http://dx.doi.org/10.1177/0363546506290186
4. Chmielewski TL. Asymmetrical lower extremity loading after ACL reconstruction: more than meets the eye. *J Orthop Sports Phys Ther*. 2011;41:374-376. http://dx.doi.org/10.2519/jospt.2011.0104
5. Chorba RS, Chorba DJ, Bouillon LE, Overmyer CA, Landis JA. Use of a functional movement screening tool to determine injury risk in female collegiate athletes. *N Am J Sports Phys Ther*. 2010;5:47-54.
6. Eastlack ME, Axe MJ, Snyder-Mackler L. Laxity, instability, and functional outcome after ACL injury: copers versus noncopers. *Med Sci Sports Exerc*. 1999;31:210-215.
7. Faude O, Junge A, Kindermann W, Dvorak J. Risk factors for injuries in elite female soccer players. *Br J Sports Med*. 2006;40:785-790. http://dx.doi.org/10.1136/bjsm.2006.027540
8. Fitzgerald GK, Lephart SM, Hwang JH, Wainner RS. Hop tests as predictors of dynamic knee stability. *J Orthop Sports Phys Ther*. 2001;31:588-597.
9. Hitchcock KM, Millard-Stafford ML, Phillips JM, Snow TK. Metabolic and thermoregulatory responses to a simulated American football practice in the heat. *J Strength Cond Res*. 2007;21:710-717. http://dx.doi.org/10.1519/R-20916.1
10. Iosia MF, Bishop PA. Analysis of exercise-to-rest ratios during Division IA televised football competition. *J Strength Cond Res*. 2008;22:332-340. http://dx.doi.org/10.1519/JSC.0b013e31816607b0
11. Johnston RB, 3rd, Howard ME, Cawley PW, Losse GM. Effect of lower extremity muscular fatigue on motor control performance. *Med Sci Sports Exerc*. 1998;30:1703-1707.
12. Keays SL, Bullock-Saxton JE, Newcombe P, Keays AC. The relationship between knee strength and functional stability before and after anterior cruciate ligament reconstruction. *J Orthop Res*. 2003;21:231-237. http://dx.doi.org/10.1016/S0736-0266(02)00160-2
13. Kiesel K, Plisky PJ, Voight ML. Can serious injury in professional football be predicted by a preseason functional movement screen? *N Am J Sports Phys Ther*. 2007;2:147-158.
14. Knapik JJ, Bauman CL, Jones BH, Harris JM, Vaughan L. Preseason strength and flexibility imbalances associated with athletic injuries in female collegiate athletes. *Am J Sports Med*. 1991;19:76-81.
15. Kvist J. Rehabilitation following anterior cruciate ligament injury: current recommendations for sports participation. *Sports Med*. 2004;34:269-280.
16. Lephart SM, Perrin DH, Fu FH, Gieck JH, McCue FC, Irrgang JJ. Relationship between selected physical characteristics and functional capacity in the anterior cruciate ligament-insufficient athlete. *J Orthop Sports Phys Ther*. 1992;16:174-181.
17. Myer GD, Paterno MV, Ford KR, Quatman CE, Hewett TE. Rehabilitation after anterior cruciate ligament reconstruction: criteria-based progression through the return-to-sport phase. *J Orthop Sports Phys Ther*. 2006;36:385-402. http://dx.doi.org/10.2519/jospt.2006.2222
18. Myer GD, Schmitt LC, Brent JL, et al. Utilization of modified NFL combine testing to identify functional deficits in athletes following ACL reconstruction. *J Orthop Sports Phys Ther*. 2011;41:377-387. http://dx.doi.org/10.2519/jospt.2011.3547
19. Nadler SF, Malanga GA, Feinberg JH, Prybicien M, Stitik TP, DePrince M. Relationship between hip muscle imbalance and occurrence of low back pain in collegiate athletes: a prospective study. *Am J Phys Med Rehabil*. 2001;80:572-577.
20. Neeb TB, Aufdemkampe G, Wagener JH, Mastenbroek L. Assessing anterior cruciate ligament injuries: the association and differential value of questionnaires, clinical tests, and functional tests. *J Orthop Sports Phys Ther*. 1997;26:324-331.
21. Noyes FR, Barber SD, Mangine RE. Abnormal lower limb symmetry determined by function hop tests after anterior cruciate ligament rupture. *Am*

Journal of Orthopaedic & Sports Physical Therapy®
Downloaded from www.jospt.org at on October 18, 2016. For personal use only. No other uses without permission.
Copyright © 2012 Journal of Orthopaedic & Sports Physical Therapy®. All rights reserved.

J Sports Med. 1991;19:513-518.
22. Noyes FR, Berrios-Torres S, Barber-Westin SD, Heckmann TP. Prevention of permanent arthrofibrosis after anterior cruciate ligament reconstruction alone or combined with associated procedures: a prospective study in 443 knees. *Knee Surg Sports Traumatol Arthrosc*. 2000;8:196-206.
23. Orchard J, Seward H, McGivern J, Hood S. Intrinsic and extrinsic risk factors for anterior cruciate ligament injury in Australian footballers. *Am J Sports Med*. 2001;29:196-200.
24. Orchard JW. Intrinsic and extrinsic risk factors for muscle strains in Australian football. *Am J Sports Med*. 2001;29:300-303.
25. Petschnig R, Baron R, Albrecht M. The relationship between isokinetic quadriceps strength test and hop tests for distance and one-legged vertical jump test following anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther*. 1998;28:23-31.
26. Plisky PJ, Rauh MJ, Kaminski TW, Underwood FB. Star Excursion Balance Test as a predictor of lower extremity injury in high school basketball players. *J Orthop Sports Phys Ther*. 2006;36:911-919. http://dx.doi.org/10.2519/jospt.2006.2244
27. Poliquin C, Patterson P. Classification of strength qualities. *Strength Cond J*. 1989;11:48-52.
28. Powers CM. The influence of abnormal hip mechanics on knee injury: a biomechanical perspective. *J Orthop Sports Phys Ther*. 2010;40:42-51. http://dx.doi.org/10.2519/jospt.2010.3337
29. Raleigh MF, McFadden DP, Deuster PA, et al. Functional movement screening: a novel tool for injury risk stratification of warfighters. *American College of Sports Medicine 57th Annual Meeting, June 2-5, 2010*. Baltimore, MD: American College of Sports Medicine; 2010.
30. Risberg MA, Holm I, Tjomsland O, Ljunggren E, Ekeland A. Prospective study of changes in impairments and disabilities after anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther*. 1999;29:400-412.
31. Sernert N, Kartus J, Kohler K, et al. Analysis of subjective, objective and functional examination tests after anterior cruciate ligament reconstruction. A follow-up of 527 patients. *Knee Surg Sports Traumatol Arthrosc*. 1999;7:160-165.
32. Sgaglione NA, Warren RF, Wickiewicz TL, Gold DA, Panariello RA. Primary repair with semitendinosus tendon augmentation of acute anterior cruciate ligament injuries. *Am J Sports Med*. 1990;18:64-73.
33. Siff MC, Verkhoshansky YV. *Supertraining: Strength Training for Sporting Excellence*. Johannesburg, South Africa: University of the Witwatersrand; 1998.
34. Tegner Y, Lysholm J, Lysholm M, Gillquist J. A performance test to monitor rehabilitation and evaluate anterior cruciate ligament injuries. *Am J Sports Med*. 1986;14:156-159.
35. Tyler TF, McHugh MP, Gleim GW, Nicholas SJ. Association of KT-1000 measurements with clinical tests of knee stability 1 year following anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther*. 1999;29:540-545.
36. Weber MD, Woodall WR. Knee rehabilitation. In: Andrews JR, Harrelson GL, Wilk KE, eds. *Physical Rehabilitation of the Injured Athlete*. 3rd ed. Philadelphia, PA: Saunders; 2004:377-428.
37. Wilk KE, Romaniello WT, Soscia SM, Arrigo CA, Andrews JR. The relationship between subjective knee scores, isokinetic testing, and functional testing in the ACL-reconstructed knee. *J Orthop Sports Phys Ther*. 1994;20:60-73.
38. Zazulak BT, Hewett TE, Reeves NP, Goldberg B, Cholewicki J. Deficits in neuromuscular control of the trunk predict knee injury risk: a prospective biomechanical-epidemiologic study. *Am J Sports Med*. 2007;35:1123-1130. http://dx.doi.org/10.1177/0363546507301585

@ MORE INFORMATION
WWW.JOSPT.ORG

Journal of Orthopaedic & Sports Physical Therapy®
Downloaded from www.jospt.org at on October 18, 2016. For personal use only. No other uses without permission.
Copyright © 2012 Journal of Orthopaedic & Sports Physical Therapy®. All rights reserved.

SEND Letters to the Editor-in-Chief

JOSPT welcomes **letters related to professional issues or articles published in the *Journal***. The Editor-in-Chief reviews and selects letters for publication based on the topic's relevance, importance, appropriateness, and timeliness. Letters should include a summary statement of any conflict of interest, including financial support related to the issue addressed. In addition, letters are copy edited, and the correspondent is not typically sent a version to approve. Letters to the Editor-in-Chief should be sent electronically to **jospt@jospt.org**. Authors of the relevant manuscript are given the opportunity to respond to the content of the letter.